

Autologous Cellular Immunotherapy Treatment

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Tracking Information

Publication Number

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Manual Section Number

110.22

Manual Section Title

Autologous Cellular Immunotherapy Treatment

Version Number

1

Effective Date of this Version

06/30/2011

Implementation Date

08/08/2011

Description Information

Benefit Category

Outpatient Hospital Services Incident to a Physician's Service
Physicians' Services

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

Prostate cancer is the most common non-cutaneous cancer in men in the United States. In 2009, an estimated 192,280 new cases of prostate cancer were diagnosed and an estimated 27,360 deaths were reported. The National Cancer Institute states that prostate cancer is predominantly a cancer of older men; the median age at diagnosis is 72 years. Once the patient has castration-resistant, metastatic prostate cancer the median survival is generally less than two years.

In 2010 the Food and Drug Administration (FDA) approved sipuleucel-T (PROVENGE®; APC8015), for patients with castration-resistant, metastatic prostate cancer. The posited mechanism of action, immunotherapy, is different from that of anti-cancer chemotherapy such as docetaxel. This is the first immunotherapy for prostate cancer to receive FDA approval.

The goal of immunotherapy is to stimulate the body's natural defenses (such as the white blood cells called dendritic cells, T-lymphocytes and mononuclear cells) in a specific manner so that they attack and destroy, or at least prevent, the proliferation of cancer cells. Specificity is attained by intentionally exposing a patient's white blood cells to a particular protein (called an antigen) associated with the prostate cancer. This exposure "trains" the white blood cells to target and attack the prostate cancer cells. Clinically, this is expected to result in a decrease in the size and/or number of cancer sites, an increase in the time to cancer progression, and/or an increase in survival of the patient.

Sipuleucel-T differs from other infused anti-cancer therapies. Most such anti-cancer therapies are manufactured and sold by a biopharmaceutical company and then purchased by and dispensed from a pharmacy. In contrast, once the decision is made to treat with sipuleucel-T, a multi-step process is used to produce sipuleucel-T. Sipuleucel-T is made individually for each patient with his own white blood cells. The patient's white blood cells are removed via a procedure called leukapheresis. In a laboratory the white blood cells are exposed to PA2024, which is a molecule created by linking prostatic acid phosphatase (PAP) with granulocyte/macrophage-colony stimulating factor (GM-CSF). PAP is an antigen specifically associated with prostate cancer cells; GM-CSF is a protein that targets a receptor on the surface of white blood cells. Hence, PAP serves to externally manipulate the immunological functioning of the patient's white blood cells while GM-CSF serves to stimulate the white blood cells into action. As noted in the FDA's clinical review, each dose of sipuleucel-T contains a minimum of 40 million treated white blood cells, however there is "high inherent variability" in the yield of sipuleucel-T from leukapheresis to leukapheresis in the same patient as well as from patient to patient. The treated white blood cells are then infused back into the same patient. The FDA-approved dosing regimen is three doses with each dose administered two weeks apart. The total treatment period is four weeks.

Indications and Limitations of Coverage

B. Nationally Covered Indications

Effective for services performed on or after June 30, 2011, The Centers for Medicare and Medicaid Services (CMS) proposes that the evidence is adequate to conclude that the use of autologous cellular immunotherapy treatment - sipuleucel-T; PROVENGE® improves health outcomes for Medicare beneficiaries with asymptomatic or minimally symptomatic metastatic castrate-resistant (hormone refractory) prostate cancer, and thus is reasonable and necessary for this on-label indication under 1862(a)(1)(A) of the Social Security Act.

C. Nationally Non-Covered Indications

N/A

D. Other

Effective for services performed on or after June 30, 2011, coverage of all off-label uses of autologous cellular immunotherapy treatment – sipuleucel-T; PROVENGE® for the treatment of prostate cancer is left to the discretion of the local Medicare Administrative Contractors.

(NCD last reviewed June 2011.)

Claims Processing Instructions

[TN 2254 \(Medicare Claims Processing\)](#) 

[TN 2339 \(Medicare Claims Processing\)](#) 

[TN 2380 \(Medicare Claims Processing\)](#) 

Transmittal Information

Transmittal Number

140

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/downloads/R140NCD.pdf> 

Revision History

01/2012 - Transmittal 136, dated November 2, 2011, is being rescinded and replaced with Transmittal 140 dated January 6, 2012. The changes include additional clarification under the Policy section regarding payment for administration of PROVENGE®, deletion of original business requirements (BRs) 8, 8.1, 9, and 9.1 relating to Common Working File frequency edits, and a date change from April 2, 2012, to July 2, 2012, in current BR 9 as it relates to Pub. 100-04 (BR). Language in current BR 7 has been revised to align with the policy changes made that allow separate payment for the cost of administration. All other information remains the same [\(TN 140\)](#)  (CR7431).

11/2011 - Transmittal 133, dated July 8, 2011, is being rescinded and replaced with Transmittal 136, dated November 2, 2011. The changes include additional clarification under the Policy section regarding the payment for administration and deletion of the original business requirements (BRs) 8, 8.1, 9 and 9.1, including referencing the Common Working File as it relates to Pub. 100-04 (BR). In addition, the language in the current business requirement (BR) 7 has been revised to allow a separate payment for the cost of administration. All other information remains the same [\(TN 136\)](#)  (CR7431).

07/2011 - Effective for services performed on or after June 30, 2011, The Centers for Medicare and Medicaid Services (CMS) proposes that the evidence is adequate to conclude that the use of autologous cellular immunotherapy treatment - sipuleucel-T; PROVENGE, improves health outcomes for Medicare beneficiaries with asymptomatic or minimally symptomatic metastatic castrate-resistant (hormone refractory) prostate cancer, and thus is reasonable and necessary for this on-label indication under 1862(a)(1)(A) of the Social

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Autologous Cellular Immunotherapy Treatment of Metastatic Prostate Cancer \(CAG-00422N\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
Autologous Cellular Immunotherapy Treatment	1	06/30/2011 - N/A	You are here